

Nehal Choudhary

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EDUCATION

University of California San Diego

2027

B.S. Mathematics-Computer Science, B.S. Cognitive Science-Machine Learning

GPA: 3.9/4.0

- Relevant Coursework: Design & Analysis of Algorithms, Advanced Data Structures, Mathematics of Machine Learning, Systems Programming, Probability, Linear Algebra

EXPERIENCE

AI & ML Device Engineering Intern | *EchoStar / Boost Mobile*

June 2026 - August 2026

- **Built an AI-powered Android test automation platform** using Python, FastAPI, React/TypeScript, Appium, and Claude Sonnet via Amazon Bedrock, converting plain-English telecom tests into reusable workflows and reducing manual testing effort by **80%+**.
- **Engineered an MCP-based planning and validation layer** connecting Claude Sonnet with device-control tools to generate, validate, and cache reusable automation flows across Android devices and chipset variations.
- **Scaled concurrent scheduling and observability across 32+ phones and 110+ test scenarios**, adding conflict handling, disconnect recovery, and step-level diagnostics; uncovered a critical **Samsung-to-Samsung RTT defect** and earned **2nd place** at EchoStar's AI Demo Fair.

Undergraduate Researcher, Multimodal AI | *UCSD Cognitive Development Lab*

December 2025 - June 2026

- **Developed a reusable Python evaluation pipeline** for Whisper and YOLOv8 across **12 infant-caregiver recordings**, standardizing preprocessing, metric computation, experiment tracking, and result logging.
- **Benchmarked Whisper on 2,665 multilingual speech segments and YOLOv8 across 10 video sessions**, identifying language-specific ASR failures and achieving **0.912 mean face-detection precision**.

LLM Evaluation Intern | *Netcracker Technology*

July 2025 - September 2025

- **Built an automated LLM evaluation framework** covering **120+ test scenarios across 6 reliability metrics**, replacing manual prompt reviews with repeatable benchmarking and regression testing.
- **Ran A/B experiments across 20+ prompt variants** to diagnose hallucination, scope drift, and schema instability, translating results into prompt-engineering and model-selection recommendations.

PROJECTS

ClassGraph, 1st Place *React · Go · Python · D3*

- **Modeled 1,000+ UCSD courses and 100+ prerequisite chains as directed graphs**, implementing recursive prerequisite traversal and validation for complex dependency structures.
- **Built academic-history PDF uploads, course search, OR-grouped requirements, and duplicate-node handling**, improving prerequisite-graph correctness and usability.

Create Similar Playlist *Python · Streamlit · Last.fm API · Spotify API*

- **Built a hybrid music recommendation pipeline** combining Last.fm collaborative similarity, multi-seed reciprocal-rank fusion, and **TF-IDF/cosine tag similarity** to rank candidates across 1-8 seed tracks.
- **Implemented MMR-style diversity reranking and Spotify integrations** for playlist import/export, track previews, matching, and configurable per-artist constraints in a deployed Streamlit application.

WildScan, 2nd Place *Python · PyTorch · FastAPI · React*

- **Built a multimodal React/FastAPI application** routing image and audio uploads across specialized wildlife classifiers and returning species identifications with LLM-generated threat guidance.
- **Trained and evaluated five CNN classifiers** using ResNet, EfficientNet, ConvNeXt, and audio CNN architectures, achieving **71-90% test accuracy** and **100-300 ms image latency**.

SKILLS AND TOOLS

Languages: Python, C++, Java, SQL, JavaScript/TypeScript, Go

ML & AI: PyTorch, scikit-learn, Model Evaluation, CNNs, LSTM/GRU/TCN, YOLOv8, Whisper, Claude Sonnet, Amazon Bedrock

Frameworks & APIs: FastAPI, React, Streamlit, REST APIs, pandas, NumPy

Infrastructure: Docker, AWS, MongoDB, Git, Appium, ADB, Model Context Protocol (MCP), Hugging Face Spaces